AP CALCULUS AB

Unit 1 • Lesson 1.3 notes

Lesson 1.3 • Estimating Limit Values from Graphs

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

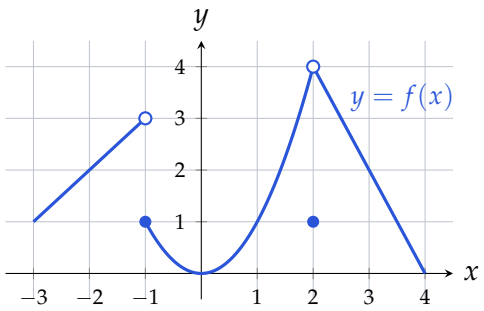
Learning goals

- Read one-sided and two-sided limits from a graph.
- Recognize graphical reasons a limit fails to exist.

Vocabulary. open circle, filled dot, jump, vertical asymptote, oscillation.

1. Follow the graph toward $x = c$

Trace the graph from the left toward $x = c$ and note the height it approaches; repeat from the right. Open and filled circles only tell you the value $f(c)$.



Example 1 • Using the graph of f above

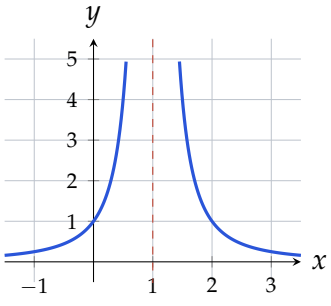
Solution.

$x = -3$ (endpoint):	$\lim_{x\to -3^+} f(x) = 1$	(only a right-hand limit)
$x = -1$:	left 3, right 1	limit does not exist; $f(-1) = 1$
$x = 0$:	$\lim_{x\to 0} f(x) = 0$	$f(0) = 0$
$x = 2$:	left 4, right 4	$\lim_{x\to 2} f(x) = 4$, but $f(2) = 1$
$x = 3$:	$\lim_{x\to 3} f(x) = 2$	$f(3) = 2$

2. Three ways a limit can fail

- **Jump**: different one-sided limits.
- **Unbounded**: the graph rises or falls without bound (vertical asymptote).
- **Oscillation**: the graph keeps oscillating, as $\sin\frac{1}{x}$ does near 0.

Example 2 • Unbounded



Solution. Near $x = 1$ the graph rises without bound on both sides: $\lim_{x\to 1} \frac{1}{(x-1)^2} = \infty$. The limit does not exist as a real number.

Common mistakes

- Reading the filled dot as the limit.
- Forgetting that at an endpoint of the domain only a one-sided limit makes sense.

Quick check

Use the graph of f above. 1. $\lim_{x\to -1^-} f(x)$? 2. $\lim_{x\to 1} f(x)$? 3. Is $\lim_{x\to 2} f(x) = f(2)$?

Answers. 1. 3. 2. 1. 3. No: $4 \neq 1$.